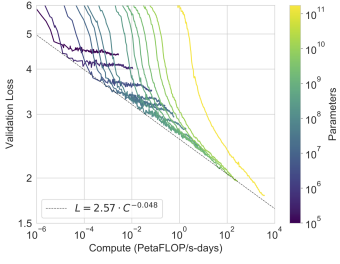
Deep Learning for NLP

M Pre-Training
Scaling laws



Learning goals

- Understand scaling laws

FROM COMPUTE BUDGET TO MODEL DESIGN I

Training compute is approximately

$$C \approx 6PD,$$

where P is the number of parameters and D the number of training tokens.
For a fixed compute budget C,

$$PD \approx \text{constant}.$$

We can therefore spend our compute on

- a **larger model** trained on fewer tokens, or
- a **smaller model** trained on more tokens.

Which choice gives the best model?

Scaling laws help us answer this question.

WHAT ARE SCALING LAWS? I

▸ Kaplan et al., 2020

Language-model loss changes **predictably with scale**.
Key scaling variables are

- model size P,
- training data D,
- training compute C.

Over large ranges, loss approximately follows power laws, e.g.,

$$L(D) \propto D^{-\alpha}.$$

Increasing only one factor eventually gives **diminishing returns**.

WHY WERE SCALING LAWS IMPORTANT? I

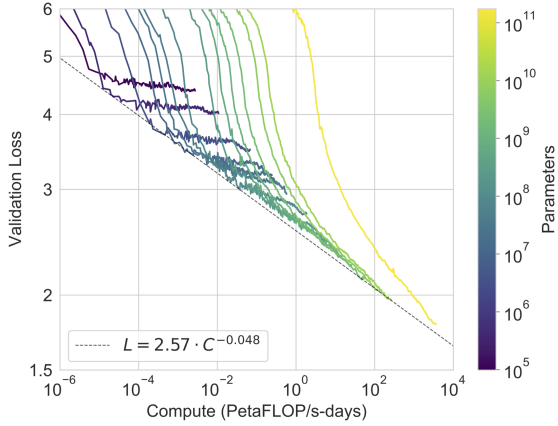
The results of Kaplan et al. were influential because they suggested that progress from scaling was remarkably **predictable**.

- Larger models and more data reliably reduced loss.
- Trends observed at small scale could be extrapolated to much larger training runs.
- Model **scale** mattered more than many details of model shape, such as depth vs. width.
- Larger models were more sample-efficient: they achieved a given loss with fewer training examples.

Scaling became a quantitative tool for planning LLM training.

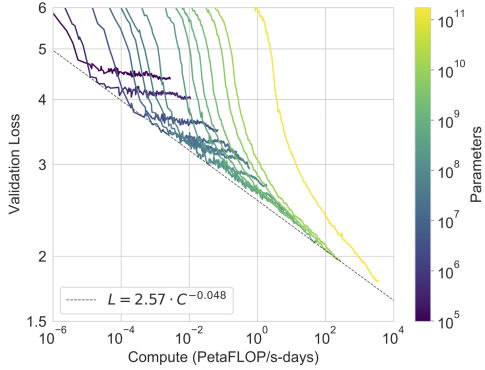
▸ Kaplan et al., 2020

WHAT DOES A POWER LAW LOOK LIKE? I



Validation loss decreases smoothly as training compute increases. The approximately straight trend on log–log axes indicates a **power law**. ▸ Brown et al., 2020

SCALING ONE THING IS NOT ENOUGH I



For a **fixed model size**, more training initially reduces loss strongly, but improvements become smaller. Likewise, **increasing model size** without enough training data eventually gives diminishing returns.

Good scaling requires balancing model size and training data.

A SCALING LAW FOR LOSS I

Loss can be modeled as a function of both model size and training data:

$$L(P, D) = 1.61 + \frac{406.4}{P^{0.34}} + \frac{410.7}{D^{0.28}}$$

where

- P is the number of parameters,
- D is the number of training tokens,
- $L(P, D)$ is cross-entropy loss on new text.

Increasing either P or D reduces its contribution to the loss, but with diminishing returns.

▸ Hoffmann et al., 2022

HOW SHOULD WE SPEND A FIXED COMPUTE BUDGET? I

Recall:

$$C \approx 6PD.$$

Suppose two training runs have the same compute budget:

	P	D
A	10^9	10^9
B	10^8	10^{10}

Both have the same product PD.
For this exercise, use the simplified scaling law

$$L(P, D) = 2 + \frac{1000}{P^{1/3}} + \frac{1000}{D^{1/4}}.$$

Which configuration should achieve lower loss?

COMPUTE-OPTIMAL TRAINING I

Given a fixed compute budget C, choose model size P and training data D to minimize loss:

$$\text{minimize } L(P, D) \quad \text{subject to} \quad 6PD = C$$

▸ Hoffmann et al., 2022

Hoffmann et al. estimated the optimal trade-off empirically using hundreds of training runs with different model sizes and dataset sizes. Their conclusion:

For a fixed compute budget, models should generally be smaller and trained on more data than previously assumed.

CHINCHILLA: SMALLER MODEL, MORE DATA I

▸ Hoffmann et al., 2022

Chinchilla was designed to test the compute-optimal prediction.

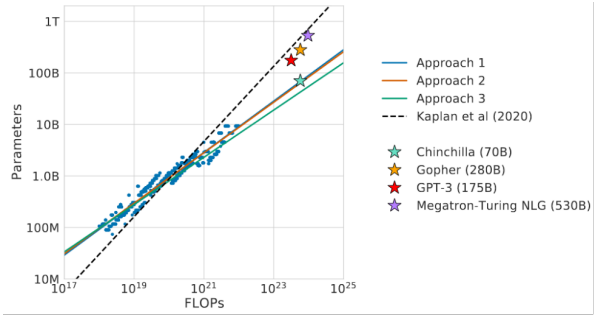
- **Chinchilla:** 70B parameters, 1.4T training tokens
- About **4**$\times$ **fewer parameters** than Gopher
- About **4**$\times$ **more training tokens**

Despite being much smaller, Chinchilla

- outperformed Gopher,
- required less memory for inference,
- had lower inference cost.

Bigger is not always better: training longer can be a better use of compute.

WERE PREVIOUS LLMS TOO LARGE? I



The Chinchilla scaling law predicted that, for a given compute budget, many earlier LLMs used ▸ Hoffmann et al., 2022

- too many parameters,
- too few training tokens.

Scaling laws shifted compute from model size to data.

MAIN TAKEAWAY I

Scaling laws make LLM training a **resource-allocation problem**.
For a fixed compute budget,

$$C \approx 6PD,$$

so more parameters mean fewer training tokens, and vice versa.
Empirical scaling laws tell us how to balance both:

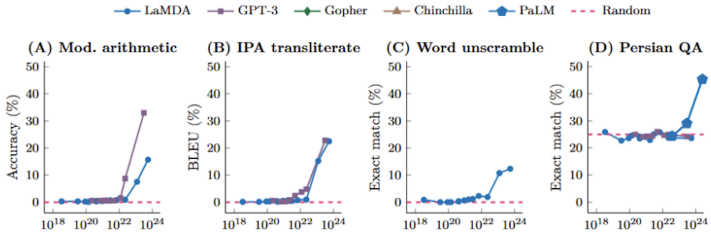
choose P and D jointly, not independently

Chinchilla showed that many earlier LLMs were **too large and undertrained** for their compute budget.
A smaller, better-trained model can also reduce **inference memory and cost**.

DO NEW ABILITIES EMERGE WITH SCALE? I

Some downstream abilities appear only above a certain model scale.

- Small models perform near chance on a task.
- Larger models suddenly achieve substantial performance.
- This has been called an **emergent ability**.

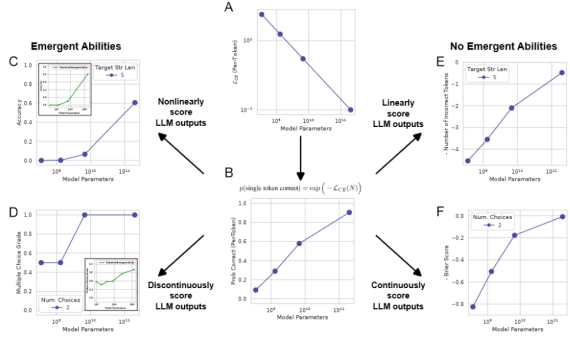


▸ Wei et al., 2022: Emergent Abilities of Large Language Models

IS EMERGENCE AN ARTIFACT OF THE METRIC? I

▸ Schaeffer et al., 2023 argue that apparent emergence can arise from how performance is measured: **A sudden benchmark jump does not imply a sudden change in the underlying model.**

- Underlying model predictions may improve smoothly with scale.
- Discontinuous metrics such as exact match can turn smooth improvement into an abrupt jump.



PRETRAINING DATA ARE MIXTURES I

The number of tokens D does not describe the **kind of data**.
Example: Dolma combines several sources

Source	Doc Type	UTF-8 bytes *(GB)*	Documents *(millions)*	Unicode words *(billions)*	Llama tokens *(billions)*
Common Crawl	web pages	9,812	3,734	1,928	2,479
GitHub	code	1,043	210	260	411
Reddit	social media	339	377	72	89
Semantic Scholar	papers	268	38.8	50	70
Project Gutenberg	books	20.4	0.056	4.0	6.0
Wikipedia, Wikibooks	encyclopedic	16.2	6.2	3.7	4.3
Total		**11,519**	**4,367**	**2,318**	**3,059**

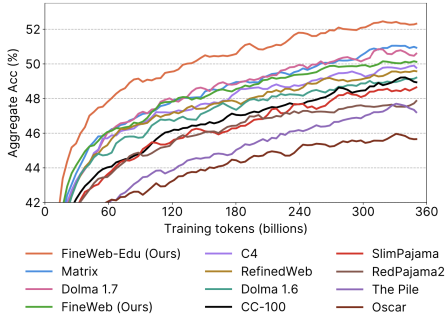
▸ Soldaini et al., 2024: Dolma

The composition of D is itself a design choice.

WHICH DATA SHOULD WE KEEP? I

Example: FineWeb

- FineWeb contains about **15T filtered web tokens**.
- FineWeb-Edu keeps about **1.3T tokens** with high predicted educational value (“useful for teaching”, judged by an LLM-derived classifier).
- Models trained on FineWeb-Edu perform better on several knowledge and reasoning benchmarks.

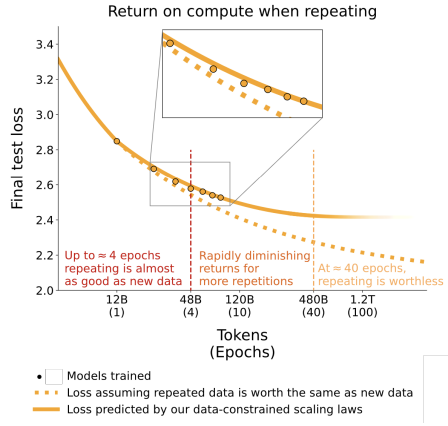


▸ Penedo et al., 2024: FineWeb

WHAT IF WE RUN OUT OF NEW DATA? I

If high-quality unique data are limited, one option is to **repeat existing data**.
Muennighoff et al. find that

- a few repetitions almost as useful as fresh unique data,
- but additional repetitions increasingly less useful.



▸ Muennighoff et al., 2023

OLMO 2: DETAILS MAKE SCALING WORK I

Scaling laws say how to allocate compute. The open-source OLMo 2 LLM shows what is needed so the training run actually remains **stable**.

▸ OLMo Team, 2025: 2 OLMo 2 Furious

Choice	OLMo 2 takeaway
Repeated text	Filter/mask long repeated *n*-grams; reduces gradient spikes.
Initialization	Use $\mathcal{N}(0, 0.02)$ for all parameters; fewer spikes and better activation/gradient scaling.
Norm placement	RMSNorm on attention/MLP outputs, not inputs.
Attention logits	QK-norm prevents very large attention logits.
Logit scale	Add z-loss to keep output logits from growing too large.
Optimizer details	Lower AdamW ϵ; do not apply weight decay to embeddings.

OLMO 2: SHORT MID-TRAINING, LARGE GAINS I

After long general pretraining, OLMo 2 performs a much shorter **targeted mid-training stage** on filtered higher-quality data.

Model	Training tokens	Avg.
7B	4T + **150B***	53.0 → **62.9**
13B	5T + **600B***	58.9 → **68.3**
32B	6T + **600B***	66.3 → **73.3**

***Total mid-training compute across several continuations:** 7B: 3 × 50B; 13B/32B: 3 × 100B + 300B. All runs start from the same pretrained checkpoint and their final checkpoints are **averaged (model soup)**.

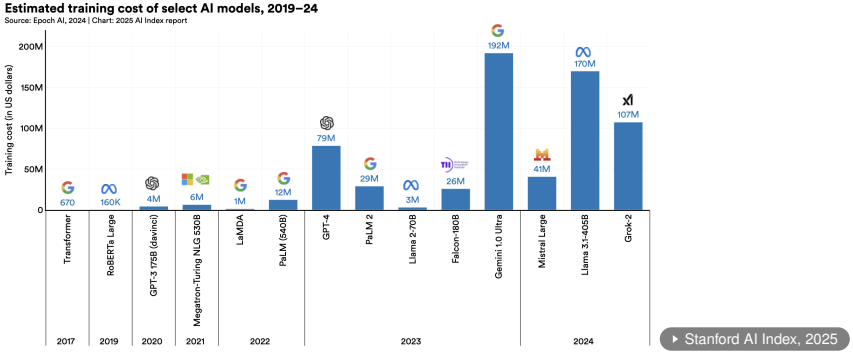
- Mid-training switches to the curated **DoImino-Mix-1124** mixture and anneals the learning rate.
- Despite requiring far fewer tokens than general pretraining, it yields **large improvements**.

▸ OLMo Team, 2025: 2 OLMo 2 Furious

SCALING HAS A FINANCIAL COST I

Estimates for training frontier models, based on cloud-compute rental prices (not total development costs):

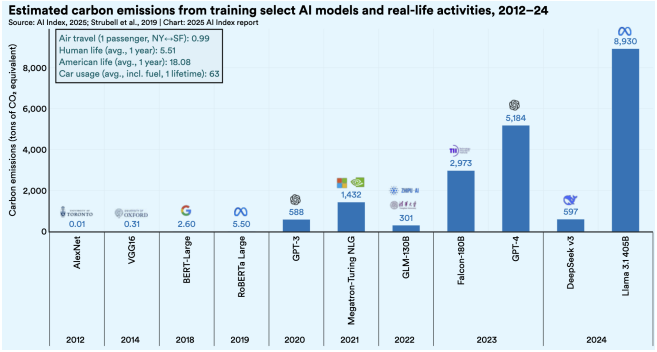
- Estimated training compute cost: **GPT-3: $4M**, **GPT-4: $79M**, **Llama 3.1-405B: $170M**.
- Experiments, tuning, and failed runs add further cost.



▸ Stanford AI Index, 2025

ENVIRONMENTAL COSTS OF TRAINING I

- Estimated training emissions: **GPT-3: 588 t**, **GPT-4: 5,184 t**, **Llama 3.1-405B: 8,930 t CO_2e**.
- Emissions depend on hardware efficiency, datacenter efficiency, and the carbon intensity of electricity.
- Patterson et al. observed about **5–10**$\times$ differences across locations. ▸ Patterson et al., 2021



▸ Stanford AI Index, 2025